Section 7 concludes, while all proofs are provided in the Appendix.

2 Model

There is an underlying state of the world $\theta \in \{L, R\}$, for example, corresponding to whether the left-wing or the right-wing candidate is more qualified for political office. Agents have heterogeneous prior (ideological) beliefs about θ, and agent i's prior that $\theta = R$ is denoted by b_i with an *ex ante* distribution $H_i(\cdot)$, which may or may not be the same across agents.

Sharing Network. We assume there are N agents in the population, who share a news item according to a *sharing network* defined by matrix $\mathbf{P}$ of link probabilities:

$$\mathbf{P} \equiv \begin{pmatrix} 0 & p_{12} & \cdots & p_{1N} \\ p_{21} & 0 & \cdots & p_{2N} \\ \cdots & \cdots & \cdots & \cdots \\ p_{N1} & p_{N2} & \cdots & 0 \end{pmatrix}$$

where p_{ij} is the probability that agent i has a link to agent j. We define agent i's neighborhood $\mathcal{N}_i$ as the set of agents attached to her with an outgoing link, and denote her degree or the size of her neighborhood by $|\mathcal{N}_i|$. The sharing network reflects both an individual's social circle and the algorithms the platform uses for promoting shared content. The news item in question could be a news article or a post by one of the users, and throughout we refer to it as an "article".

Misinformation and News Generation. Each article has a three-dimensional type (r, m, ν). Here, $r \in [0, 1]$ indicates the *reliability* of the news, and $m \in \{L, R\}$ is the *message*, which corresponds to the article's viewpoint, for example, whether it argues for a left-wing or right-wing idea. Finally, ν is the article's *veracity*, which can either be $\mathcal{T}$, to indicate the article is truthful, or $\mathcal{M}$, to indicate the article contains *misinformation*.[8]

We assume that, at the beginning of the game, the type vector (r, ν, m) is drawn according to the following i.i.d. process:

(i) The article's reliability $r \in [0, 1]$ is drawn from a continuous distribution F with density f.

(ii) The veracity of the article is $\nu = \mathcal{T}$ (contains truthful content) with probability $\phi(r)$ or is $\nu = \mathcal{M}$ (contains misinformation) with probability $1 - \phi(r)$. We assume that ϕ is increasing and differentiable in r, and satisfies $\phi(0) = 0$ and $\phi(1) = 1$, so that the least reliable article always contains misinformation, and as the degree of reliability increases, the likelihood of misinformation monotonically declines and reaches zero.

[8]Our focus in this paper is on misinformation, interpreted as items containing misleading information or arguments that can influence (a subset of) the public. Articles containing misinformation are in practice much more numerous than those that can be classified as "fake news", which explicitly propagate demonstrably false information (e.g., Egelhofer and Lecheler (2019), Allen et al. (2020), Guess et al. (2019), Grinberg et al. (2019)). For example, according to this definition a news item that favorably describes a report denying climate change, without putting this in the context of hundreds of other reports reaching the opposite conclusion or mentioning the criticisms that it has received from experts, contains misinformation.